

CARRY OTTIMIZZATO (LOOKAHEAD):

$$g_i = A_i \cdot B_i$$

$$p_i = A_i + B_i$$

$$C_i = g_i + p_i \cdot C_0$$

$$C_1 = g_0 + p_0 \cdot C_0$$

$$C_2 = g_1 + p_1 \cdot C_1 = g_1 + p_1 g_0 + p_1 p_0 C_0$$

$$C_3 = g_2 + p_2 \cdot C_2 = g_2 + p_2 g_1 + p_2 p_1 g_0 + p_2 p_1 p_0 C_0$$

$$C_4 = g_3 + p_3 \cdot C_3 = g_3 + p_3 g_2 + p_3 p_2 g_1 + p_3 p_2 p_1 g_0 + p_3 p_2 p_1 p_0 C_0$$

$$C_1 = \underbrace{g_0 + p_0 \cdot C_0}_G = C_4$$

$$C_1 = G_0 + p_0 \cdot C_0$$

$$C_2 = G_1 + p_1 \cdot C_1 = G_1 + p_1 G_0 + p_1 p_0 C_0$$

$$C_3 = G_2 + p_2 \cdot C_2 = G_2 + p_2 G_1 + p_2 p_1 G_0 + p_2 p_1 p_0 C_0$$

$$C_4 = G_3 + p_3 \cdot C_3 = G_3 + p_3 G_2 + p_3 p_2 G_1 + p_3 p_2 p_1 G_0 + p_3 p_2 p_1 p_0 C_0$$

$$C_1 = C_4 \quad C_2 = C_8 \quad C_3 = C_{12} \quad C_4 = C_{16}$$

$$G_1 = g_3 + p_3 g_2 + p_3 p_2 g_1 + p_3 p_2 p_1 g_0$$

$$G_2 = g_7 + p_7 g_6 + p_7 p_6 g_5 + p_7 p_6 p_5 g_4$$

$$G_3 = g_{11} + p_{11} g_{10} + p_{11} p_{10} g_9 + p_{11} p_{10} p_9 g_8$$

$$G_4 = g_{15} + p_{15} g_{14} + p_{15} p_{14} g_{13} + p_{15} p_{14} p_{13} g_{12}$$

$$P_i = P(4i) + P(4i+1) + P(4i+2) + P(4i+3) \quad i=0 \dots 3$$

	3	2	1	0	3	2	1	0	3	2	1	0	3	2	1	0
A	0	0	0	1	1	0	1	0	0	0	1	1	0	0	1	1
B	1	1	1	0	0	1	0	1	1	1	1	0	1	0	1	1
g_i	0	0	0	0	0	0	0	0	0	0	1	0	0	0	1	1
p_i	1	1	1	1	1	1	1	1	1	1	1	1	1	0	1	1
G_i	3				2				1				0			
P_i	0				0				1				0			
	1				1				1				0			

CARRY NEI SOMMATORI:

$$C_0 = 0$$

$$C_1 = 1 + \dots = 1$$

$$C_2 = 1 + \dots = 1$$

$$C_3 = 0 + 0 \cdot 1 + 0 + 0 = 0$$

$$C_4 = 0 + 0 + 0 + 0 + 0 = 0$$

$$C_4 = 0$$

$$C_5 = 0 + 1 \cdot 0 = 0$$

$$C_6 = 1 + \dots = 1$$

$$C_7 = 0 + 1 \cdot 1 + \dots = 1$$

$$C_8 = 0 + 1 \cdot 0 + 1 \cdot 1 \cdot 1 + \dots = 1$$

$$C_8 = 1$$

$$C_9 = 0 + 1 \cdot 1 = 1$$

$$C_{10} = 0 + 1 \cdot 0 + 1 \cdot 1 \cdot 1 = 1$$

$$C_{11} = 0 + 1 \cdot 0 + 1 \cdot 1 \cdot 0 + 1 \cdot 1 \cdot 1 \cdot 1 = 1$$

$$C_{12} = 0 + 1 \cdot 0 + 1 \cdot 1 \cdot 0 + 1 \cdot 1 \cdot 1 \cdot 0 + 1 \cdot 1 \cdot 1 \cdot 1 = 1$$

$$C_{12} = 1$$

$$C_{13} = 0 + 1 \cdot 1 = 1$$

$$C_{14} = 0 + 1 \cdot 0 + 1 \cdot 1 \cdot 1 = 1$$

$$C_{15} = 0 + 1 \cdot 0 + 1 \cdot 1 \cdot 0 + 1 \cdot 1 \cdot 1 \cdot 1 = 1$$

$$C_{16} = 0 + 1 \cdot 0 + 1 \cdot 1 \cdot 0 + 1 \cdot 1 \cdot 1 \cdot 0 + 1 \cdot 1 \cdot 1 \cdot 1 \cdot 1 = 1$$

CARRY:

$$C = 0$$

$$C_1 = 0 + 0 \cdot 0 = 0 = C_4$$

$$C_2 = 1 + 1 \cdot 0 + 1 \cdot 0 \cdot 0 = 1 = C_8$$

$$C_3 = 0 + 1 \cdot 1 + 1 \cdot 1 \cdot 0 + 1 \cdot 1 \cdot 0 \cdot 0 = 1 = C_{12}$$

$$C_4 = 0 + 1 \cdot 0 + 1 \cdot 1 \cdot 1 + 1 \cdot 1 \cdot 0 \cdot 0 + 1 \cdot 1 \cdot 1 \cdot 1 \cdot 0 = 1 = C_{16}$$

TEMPO (Δ TEMPO = TEMPO CHE CI METTE LA PORTA LOGICA PER DARE IL RISULTATO):

$$\left. \begin{array}{l} 1\Delta \rightarrow \text{PER } g_i \text{ E } p_i \\ 1\Delta \rightarrow \text{PER } P_i \\ 2\Delta \rightarrow \text{COMPT. I } G_i \end{array} \right\} 3\Delta + 2\Delta \rightarrow \text{PER STABILIZZARE I CARRY OUT } C_1 - C_4$$

$$+ 2\Delta \rightarrow \text{PER STABILIZZARE I CARRY-OUT INTERNI} + 2\Delta \rightarrow \text{PER STABILIZZARE GLI OUTPUT.}$$

QUINDI IN TUTTO 9Δ DI TEMPO CONTRO I 32Δ ($2\Delta \times 16 \text{ BIT}$) DI TEMPO DEL RIPPLE CARRY.